

IT INFRASTRUCTURE PROJECT

Active Directory Implementation & Network Restructure

EESI Material and Controls Corp

emcc.local | Windows Server 2016 | EMCC-ADDS01

7	28	7	8
Phases Completed	Users Created	GPOs Deployed	VLANs Managed

Prepared by
Jonel R. Laurel
MIS Supervisor
April 2026

1. Executive Summary

This document presents the complete implementation of an Active Directory (AD) domain environment for EESI Material and Controls Corp (EMCC). The project involved designing and deploying a Windows Server 2016-based AD infrastructure from the ground up, restructuring the existing network into department-based VLANs, migrating DHCP services, and establishing centralized identity and access management across all departments.

A remote-access VPN sub-project was subsequently executed to completion, delivering IKEv2 VPN via Fortigate 61E with NPS/RADIUS authentication, auto-connect scheduled tasks, and anti-tamper controls — enabling secure work-from-anywhere for all domain users.

The project was executed in seven structured phases over a controlled rollout schedule designed to ensure zero downtime to existing operations while progressively modernizing the network and IT infrastructure. All phases have been completed successfully.

2. Project Scope & Objectives

2.1 Objectives

- Deploy a centralized Active Directory domain (emcc.local) on Windows Server 2016
- Restructure the existing VLAN architecture to support AD-integrated department segmentation
- Migrate DHCP services from Cisco SG350X to Fortigate 61E and Windows Server
- Create and organize all domain user accounts across 9 departments
- Deploy Group Policy Objects (GPOs) for security, drive mapping, wallpaper, remote access, and VPN
- Enable secure IT remote management via RDP and live session shadowing across all domain laptops
- Provide secure remote access (VPN) for all domain users via IKEv2/IPsec with auto-connect

2.2 Scope

Domain Controller	Windows Server 2016 on EMCC-ADDS01 (172.30.9.3)
Network devices	Fortigate 61E, Cisco SG350X, UniFi UCK G2 Plus + 7 APs
Domain users	28 active accounts across 9 departments
AD VLANs created	4 new VLANs (101, 102, 103, 108) with dedicated subnets
Legacy VLANs	8 VLANs managed (11, 22, 31, 88 restructured; 5, 7, 9 preserved)
GPOs deployed	7 GPOs: Drive Mapping, Remote Desktop, Security Policy, USB Block, Wallpaper, Profile Picture, VPN Deployment
Shared storage	Central_EMCC 2.0 — mapped as Z: drive on all domain laptops
VPN solution	IKEv2 IPsec via Fortigate 61E + Windows built-in client + NPS/RADIUS

3. Infrastructure Overview

Component	Device / Software	Details
Domain Controller	Windows Server 2016	EMCC-ADDS01 172.30.9.3 NIC-teamed VLAN 9
Directory Services	Active Directory DS	Domain: emcc.local Forest/Domain: WinThreshold
Certificate Authority	AD CS (Enterprise Root)	emcc-EMCC-ADDS01-CA issues VPN + LDAPS certs
DNS	Windows DNS Server	AD-integrated Forwarder: 8.8.8.8 LDAPS: TCP 636
DHCP	Fortigate + Win Server	Fortigate: legacy VLANs Win Server: AD VLANs 101-108
NPS / RADIUS	Windows NPS on ADDS01	EAP-MSCHAPv2 Auth for VPN Policy: EMCC-VPN-Policy
Firewall / Gateway	Fortigate 61E	All VLAN gateways IKEv2 VPN endpoint WAN: 112.198.196.27
Core Switch	Cisco SG350X	L2 trunk only VLAN segmentation No L3 SVIs
WiFi Controller	UniFi UCK G2 Plus	172.30.5.2 7 APs managed VLAN 5

4. Project Phases & Accomplishments

Phase 1 — DC Infrastructure Preparation

Deliverable	Details	Status
Static IP configuration	172.30.9.3 / VLAN 9	COMPLETE
Server rename	Renamed to EMCC-ADDS01	COMPLETE
Roles installed	AD DS, DNS, DHCP, AD CS, NPS	COMPLETE
Domain promotion	emcc.local — WinThreshold forest mode	COMPLETE
DNS zones created	Forward + Reverse lookup zones	COMPLETE
Enterprise Root CA	emcc-EMCC-ADDS01-CA — cert authority	COMPLETE
LDAPS enabled	TCP 636 — for app integrations	COMPLETE

Phase 2 — Network Restructure

Deliverable	Details	Status
DHCP migrated	Cisco SG350X DHCP → Fortigate	COMPLETE
VLAN gateways moved	Cisco L3 SVIs → Fortigate interfaces	COMPLETE
VLANs merged/renamed	8→88, 10+20→11, 30+40→31	COMPLETE
Cisco port cleanup	PVIDs, access/trunk ports updated	COMPLETE
APs converted to static	7 APs assigned static IPs on VLAN 5	COMPLETE
Captive portal exemptions	All dept VLANs exempted on Fortigate	COMPLETE
Synology NAS	gi1/0/46 — VLAN 22, static IP preserved	COMPLETE

Phase 3 — AD VLAN Creation

Deliverable	Details	Status
VLAN 101 — MIS/Admin	10.20.1.0/24 Fortigate Admin_VLAN101	COMPLETE
VLAN 102 — Support	10.20.2.0/24 Fortigate Support_VLAN102	COMPLETE
VLAN 103 — Core	10.20.3.0/24 Fortigate Core_VLAN103	COMPLETE
VLAN 108 — Exec	10.20.8.0/24 Fortigate Exec_VLAN108	COMPLETE
Fortigate DHCP pools	Win Server DHCP for all 4 AD VLANs	COMPLETE
SSIDs created on UniFi	EMCC-ADMIN, EMCC-SUPPORT, EMCC-CORE	COMPLETE
Firewall policies	Inter-VLAN + Server VLAN9 routing	COMPLETE

Phase 4 — OU & User Structure

Deliverable	Details	Status
OU structure created	9 dept User OUs + 4 Computer OUs under OU=EMCC	COMPLETE
Service Accounts OU	OU=Service Accounts created and secured	COMPLETE
28 users provisioned	All departments — correct OU placement	COMPLETE
Department groups	Admin, Sales, HR, Finance, ECC, SCM, etc.	COMPLETE
Sales_Dept members	8 members: Aldrin, Bryan, Flordeliza, Joey, Mark, Mica, Crissy, Alejo	COMPLETE
Stale accounts removed	7 old accounts deleted (Daisy, Gab, Shiela, Alexis, Josh, Johnny, OJTMau)	COMPLETE

Phase 5 — Core GPO Deployment

Deliverable	Details	Status
EMCC - Drive Mapping	Z: → \\172.30.9.3\Central_EMCC 2.0 (User scope)	COMPLETE
EMCC - Remote Desktop	RDP + firewall + shadow + Domain Admins in RDP group	COMPLETE
EMCC-Security-Policy	Password policy, screen lock, logon banner, RPC firewall rules	COMPLETE
EMCC - USB Block	USB deny read/write/execute (standalone GPO)	COMPLETE
EMCC - Wallpaper	Domain wallpaper — placeholder; Zoho live version pending	COMPLETE
EMCC - Profile Picture	User thumbnailPhoto sync from D:\USER THUMBNAIL\	COMPLETE
Verified on test laptops	EMCC861 (flordeliza.lacdan) + EMCC182 (aldrin.rollon)	COMPLETE

Phase 6 — VPN Sub-Project (IKEv2 Remote Access)

Deliverable	Details	Status
NPS/RADIUS configured	Policy: EMCC-VPN-Policy EAP-MSCHAPv2 TCP 1812	COMPLETE
VPN cert issued	EMCC-VPN-Cert (CN=vpn.emcc.local, expires 2028-04-21)	COMPLETE
Fortigate IKEv2 tunnel	Phase1: EMCC-VPN Phase2: EMCC-VPN-P2 AES256/SHA256	COMPLETE
Firewall policies	Policy 56: VPN→Server VLAN9 Policy 57: VPN→AD VLANs	COMPLETE
Windows VPN profile	AllUserConnection IKEv2 Split tunnel DnsSuffix emcc.local	COMPLETE
VPN Deployment GPO	EMCC - VPN Deployment .bat wrapper + PS1 on NETLOGON	COMPLETE
Auto-connect task	Scheduled task: 3 triggers (Logon + NetEvent + 1-min repeat)	COMPLETE
Anti-tamper ACL	SYSTEM:(F) Admins:(F) Users:(RX) — users cannot disable	COMPLETE
End-to-end test	EMCC182 + EMCC861 connected; Central EMCC accessible remotely	COMPLETE

Phase 7 — FortiClient / Legacy GPO Audit

Deliverable	Details	Status
GPO audit	All GPOs reviewed — zero FortiClient references	COMPLETE
SYSVOL audit	Zero FortiClient scripts or files in NETLOGON/SYSVOL	COMPLETE
Legacy GPO cleanup	Old FortiClient, AnyDesk, SRP, and VPN GPOs removed	COMPLETE
Confirmed clean state	No stale entries; production environment is clean	COMPLETE

5. Final VLAN Architecture

VLAN	Name	Subnet	Gateway	DHCP Server	Status
5	AP Management	172.30.5.0/27	172.30.5.1 (FG)	Fortigate	Active
7	Voice / Avaya	172.30.7.0/24	172.30.7.1 (FG)	Fortigate	Active
9	Servers	172.30.9.0/24	172.30.9.1 (FG)	Fortigate	Active
11	HR + Finance	172.30.10.0/26	Fortigate	Fortigate	Active
22	IT Admin	—	Fortigate	Fortigate	Active
31	Sales + CCESD	—	Fortigate	Fortigate	Active
88	Executive	—	Fortigate	Fortigate	Active
101	AD MIS/Admin	10.20.1.0/24	10.20.1.1 (FG)	Windows Server	AD VLAN
102	AD Support	10.20.2.0/24	10.20.2.1 (FG)	Windows Server	AD VLAN
103	AD Core	10.20.3.0/24	10.20.3.1 (FG)	Windows Server	AD VLAN
108	AD Exec	10.20.8.0/24	10.20.8.1 (FG)	Windows Server	AD VLAN

6. Department to AD Mapping

Department	Users OU	Computers OU	VLAN	Subnet
MIS	Users\MIS	Computers\MIS	101	10.20.1.0/24
HR	Users\HR	Computers\Support	102	10.20.2.0/24
Finance	Users\Finance	Computers\Support	102	10.20.2.0/24
SCM	Users\SCM	Computers\Support	102	10.20.2.0/24
ECC	Users\ECC	Computers\Core	103	10.20.3.0/24
Sales	Users\Sales	Computers\Core	103	10.20.3.0/24
Marketing	Users\Marketing	Computers\Core	103	10.20.3.0/24
Safety	Users\Safety	Computers\Core	103	10.20.3.0/24
Exec	Users\Exec	Computers\Exec	108	10.20.8.0/24

7. VPN Architecture

Remote access is provided via Fortigate IPsec IKEv2 dial-up VPN with the Windows built-in VPN client on all domain laptops. Authentication is handled by NPS on EMCC-ADDS01 using EAP-MSCHAPv2 — enabling AD credentials for VPN login without any third-party VPN client.

7.1 Connection Parameters

Protocol	IKEv2 IPsec
Server Endpoint	112.198.196.27 (wan1, PPPoE static)
Server Certificate	EMCC-VPN-Cert CN=vpn.emcc.local Expires 2028-04-21
Authentication	EAP-MSCHAPv2 → NPS RADIUS (EMCC-ADDS01:1812)
Encryption	AES256 / SHA256 / DH Group 14 (IKE + ESP)
Client IP Pool	10.20.200.1 – 10.20.200.254
Tunnel Mode	Split tunnel — internet goes direct; internal routes via VPN
VPN Profile Type	AllUserConnection — applies to all users on the laptop

7.2 Internal Routes (VPN Client)

172.30.9.0/24	Server VLAN — DC, file share (Central_EMCC 2.0), DNS
10.20.1.0/24	AD Admin / MIS VLAN
10.20.2.0/24	AD Support VLAN (HR, Finance, SCM)
10.20.3.0/24	AD Core VLAN (Sales, ECC, Marketing, Safety)
10.20.4.0/24	AD Exec VLAN

7.3 Auto-Connect Mechanism

A scheduled task (\EMCC\EMCC VPN Auto-Connect) runs under SYSTEM with three triggers: at user logon (30s delay), on network profile change events (Event 10000, 15s delay), and on a 1-minute repeating timer. The reconnect script checks if VPN is already up or if the machine is on the office LAN (DomainAuthenticated interface) before dialing — preventing unnecessary connections in the office while ensuring remote users are always connected within ~60 seconds.

8. Pending Work Items

Deliverable	Details	Status
Profile photo sync	Server upload complete — 28 users at 1080×1080 (~93 KB). Verified on EMCC861 + EMCC182. Remaining laptops sync at next boot via GPO startup script.	IN PROGRESS
NTFS permissions	12 dept folders on Central_EMCC 2.0 — Public (Modify/no-delete) + Private (dept Full Control)	PENDING
Remaining laptop joins	<20 laptops total; EMCC861 + EMCC182 joined; follow pre-stage workflow in playbook	PENDING
Zoho CRM wallpaper	Live wallpaper GPO blocked on Zoho Client Secret — script structure ready	PENDING

Deliverable	Details	Status
Final dcdiag /v pass	Run dcdiag /v before officially closing AD project	PENDING
USB policy dedup review	EMCC-Security-Policy and EMCC - USB Block both contain USB deny — review for duplication	PENDING
VPN hard lockdown	Defer NC_RasConnect=0 GPO until all laptops have cached VPN creds	PENDING

9. Key Outcomes & Benefits

- Centralized identity management — 28 users log in with a single domain account on any domain-joined laptop
- Zero-downtime migration — all VLANs remained operational throughout the entire restructure and rollout
- Automated software and policy deployment — new laptops receive all GPOs on first domain join without IT intervention
- Remote IT management — IT admin can RDP or live-shadow any domain laptop from the DC without disrupting the user
- Secure remote access — IKEv2 VPN with AD authentication auto-connects within ~60 seconds of any network change
- Scalable infrastructure — adding a new user or laptop requires only AD account creation and a domain join
- Department-level network segmentation — each department isolated on its own VLAN with dedicated Fortigate policies
- Shared storage — Central_EMCC 2.0 mapped as Z: drive on all domain computers via GPO, accessible over VPN
- PKI foundation — Enterprise Root CA enables future certificate-based auth, LDAPS app integrations, and code signing

10. Technologies & Skills Demonstrated

Technology	Application
Windows Server 2016	AD DS, DNS, DHCP, AD CS (Enterprise CA), NPS, Group Policy, File Sharing
Active Directory	Domain promotion, OU design, user/group management, GPO architecture
Group Policy (GPO)	Drive mapping, security policy, USB control, wallpaper, VPN deployment, RDP/shadow
PowerShell	AD automation, VPN provisioning, idempotent deployment scripts, scheduled tasks
PKI / AD CS	Enterprise Root CA, VPN certificate lifecycle, LDAPS, trust chain
IKEv2 / IPsec VPN	Fortigate dial-up VPN, Windows IKEv2 client, split tunneling, auto-connect
NPS / RADIUS	EAP-MSCHAPv2, RADIUS policy, VPN authentication, AD integration
Fortigate 61E	VLAN routing, DHCP, firewall policies, captive portal, VPN endpoint, RADIUS client
Cisco SG350X	L2 switching, VLAN trunking, port PVID configuration
UniFi / APs	SSID creation, AP static IP, VLAN tagging, UCK G2 Plus management
Remote Desktop (RDP)	Admin access, live session shadowing via qwinsta/shadow

Technology	Application
TCP/IP Networking	Subnetting, inter-VLAN routing, DNS, DHCP relay, static routes

11. Certification & Signatures

I hereby certify that the Active Directory implementation, network restructure, and VPN deployment documented in this report were planned, executed, and completed by the undersigned. All configurations, policies, and infrastructure changes described herein reflect actual work performed on the EMCC production environment.

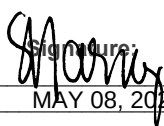
Jonel R. Laurel
MIS Supervisor
EESI Material and Controls Corp

Annie Rose M. Olivarez
HR and Admin Manager
EESI Material and Controls Corp

Mar V. Ignacio
President
EESI Material and Controls Corp

Prepared by

Signature:
Date: MAY 08, 2026

Noted by

Signature:
Date: MAY 08, 2026

Approved by

Signature:
Date: MAY 08, 2026

PROJECT COMPLETION CERTIFICATION

This certifies the successful completion of the Active Directory Implementation, Network Restructure & VPN Deployment

EESI Material and Controls Corp | emcc.local | April 2026